# Nathan Herling

MS in data science candidate · University of Arizona

Tucson, AZ 85743 · (602) 478-6137 · nth.ece.phy@gmail.com · nathanherling.com · github.com/N-Herling-Mk1

## About

Physics-trained data scientist working on machine learning for long-lived-particle searches with the ATLAS experiment at CERN. Research focus is making trained networks defensible: decorrelated training objectives, Bayesian uncertainty on the weights themselves, and independent statistical validation of the background estimate. Parallel track in detector control-system software. Preparing to continue to doctoral research in condensed matter physics, with interests spanning materials science, solid-state systems, and quantum computing.

## Education

**M.S. Data Science, University of Arizona** Summer 2025 – Dec. 2026

Graduate GPA 3.9. Capstone: FORGE, a web-hosted Bayesian posterior observatory (mentor: Prof. Kenneth Johns).

**B.S. Physics, University of Arizona** May 2024

Minor in mathematics.

**B.S. Electrical & Computer Engineering, University of Arizona** Dec. 2023

Dual electrical engineering and computer engineering tracks. Minor in computer science.

**B.S. Biology, Wayne State College, Wayne, NE**

Minor in chemistry.

## Employment

**Graduate research assistant, Department of Physics** Aug. 2025 – present

University of Arizona ATLAS high energy physics group. Appointment: 0.25 FTE fall 2025, 0.50 FTE spring 2026, 1.00 FTE summer 2026, 0.50 FTE fall 2026.

*Long-lived particle search — advisor: Prof. Kenneth Johns*

- Run-3 barrel and endcap displaced-vertex search for hidden-sector showers.
- Built the double-DisCo classifier pipeline in JAX/optax: paired 128-128 networks, five-fold out-of-fold training, cross-entropy plus a distance-correlation decorrelation penalty.
- Ran the decorrelation-strength certification campaign; located the small-coupling phase transition and fixed the production operating point at no measurable cost in discrimination.
- Added Bayesian validation across all four production networks via Hamiltonian Monte Carlo, separating data sensitivity from parameter uncertainty as distinct diagnostics.
- Built the ROOT-to-CSV feature extraction path (PyROOT, uproot, RDataFrame) now used as group tooling.

*Detector control systems — advisor: Prof. Elliott Cheu*

- Developed a simulated fleet of OPC UA servers for readout-board monitoring, with 24-hour data capture and WinCC OA panel binding.
- Building a composite sixteen-board fleet interface with per-board status, drill-down statistics, and detector-region highlighting.

**Software engineering intern, Clinisys** Summer 2023

Tucson, AZ.

- Identified and remediated security vulnerabilities in a production C# codebase.
- Built Python automation over the Veracode API with XML processing, delivering weekly vulnerability reports through Azure Pipelines.

**Research experience for undergraduates, University of Arizona** Summer 2022

Cognitive and high-frequency radio.

- Developed a reinforcement learning model for optimizing 5G virtual network slices, with supporting simulation of next-generation wireless systems.

**Undergraduate researcher, Department of Physics, University of Arizona** 2019 – 2022

Advisor: Prof. Kenneth Johns.

- Contributed C/C++ to the ATLAS muon spectrometer calibration effort, improving the accuracy of experimental data

collection.
- Wrote Python for a data-driven, model-independent machine learning approach to the long-lived particle search at the Large Hadron Collider.

## Awards and funding

- Graduate research assistantship, ATLAS high energy physics group, University of Arizona — funded appointment across four consecutive terms, 2025–2026.
- Best use of embedded intelligence — Senior Design, Electrical and Computer Engineering, University of Arizona, May 2023.
- CERN SUPER award research grant, May 2022.
- Selected for the Cognitive Radio and High Frequency Research REU, University of Arizona, 2022.

## Skills

**Languages and tools.** Python (NumPy, pandas, scikit-learn), JAX/optax, PyTorch, TensorFlow/Keras, PyROOT and uproot, C/C++, C#, Java, SQL, JavaScript (React, Three.js, HTML5 canvas), PHP, PowerShell and Unix shell, LaTeX, Git and GitHub Actions, Docker Compose, Flask, PyQt6, WinCC OA and OPC UA, Cloudflare Tunnel.

**Methods.** Bayesian inference (Hamiltonian Monte Carlo, linearized-Laplace approximation), decorrelation objectives via distance correlation, maximum-entropy density estimation and copula modeling, ABCD background estimation and closure testing, $k$-fold uncertainty decomposition, calibration and coverage, self-supervised representation learning, E(3)-equivariant networks, reinforcement learning, SCADA and data-acquisition instrumentation.

## Project portfolio

**FORGE — Bayesian posterior observatory** (forge-observatory.com)
Web-hosted instrument for interrogating a trained model's posterior: a linearized-Laplace fast path with Hamiltonian Monte Carlo as gold standard, condensed diagnostic gauges, a live prior sweep, curvature-spectrum readout, and a downloadable audit. Flask and Docker backend, self-hosted and published over a Cloudflare tunnel. Three reproductions: audio genre classification, phonon density-of-states regression with an E(3)-equivariant network, and ATLAS displaced-vertex detection.

**Maximum-entropy background estimation**
An independent second estimator for the ABCD method: per-feature maximum-entropy marginals fit on total background, then a three-rung dependence ladder from factorized product through Gaussian copula to a Gaussian-mixture copula, with samples pushed through the frozen production networks to reconstruct all four quadrant yields.

**HELIX — event representation learning**
Self-supervised event embeddings built on a Jaynes maximum-entropy reading of contrastive learning, in which the InfoNCE objective is a Boltzmann distribution and temperature is inverse thermodynamic $\beta$. Driving question: do detector events carry a learnable grammar that improves downstream classifiers?

**Senior design — adaptive health questionnaire platform**
Full-stack system (HTML, PHP, JavaScript, SQL, Python) generating health questionnaires of varying length and response type, driven by supervised machine learning. Recognized for best use of embedded intelligence.

**ROOT-to-CSV toolkit** (github.com/N-Herling-Mk1/root_to_csv)
Group conversion tooling in two layers: a scan pass classifying every branch of a ROOT file into six structural bins and emitting an editable manifest, then a convert pass producing CSV and Parquet under per-branch policy. Documented as three deployment tiers — command line on a headless server, VS Code over a browser tunnel, and a full stack adding a Flask front end — each with its own instructions and tunnel hygiene. Vetted end to end on Run-3 samples.

## Manuscripts in progress

1. **Closure-aware decorrelation for ABCD background estimation: a comparison of penalty families** (working title)

   N. Herling, [co-authors], K. Johns. Controlled ablation across three closure-penalty loss families — one drawn from the literature and two of my own construction — and two constrained-optimization trainers, judged against a preregistered criterion on closure pull. (in preparation)

2. **Maximum-entropy background estimation as an independent validator of the ABCD method in an ATLAS Run-3 long-lived-particle search** (working title)

   N. Herling, [co-authors], K. Johns. Reconstructs ABCD quadrant yields from maximum-entropy background densities sampled through the frozen classifiers, giving a closure cross-check that does not assume regional factorization. (in preparation)